

Memory.c: codes

```
GNU nano 8.7.1 memory.c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>

int global_var = 100;
int global_uninit;

static int static_global = 200;

void sample_function()
{
    printf("Code address      : %p\n", (void *)sample_function);
}

int main()
{
    int stack_var = 10;
    static int static_local = 20;

    int *heap_var = (int *)malloc(sizeof(int));
    *heap_var = 30;

    printf("Process ID          : %d\n", getpid());

    sample_function();

    printf("Global variable address : %p\n", (void *)&global_var);
    printf("Global BSS address      : %p\n", (void *)&global_uninit);
    printf("Static global address   : %p\n", (void *)&static_global);
    printf("Static local address    : %p\n", (void *)&static_local);
    printf("Heap variable address   : %p\n", (void *)heap_var);
    printf("Stack variable address  : %p\n", (void *)&stack_var);

    printf("\nProcess is running...\n");
    printf("Use /proc/%d/maps to inspect memory mappings.\n", getpid());

    sleep(120);

    free(heap_var);
}
```

Memory.c: Output

```
Process ID          : 614
Code address        : 0x59f844ddf209
Global variable address : 0x59f844de2010
Global BSS address   : 0x59f844de2020
Static global address : 0x59f844de2014
Static local address  : 0x59f844de2018
Heap variable address  : 0x59f8764d9010
Stack variable address : 0x7ffe63532d0c

Process is running...
Use /proc/614/maps to inspect memory mappings.
|
```

```
memory memory.c
uday@DESKTOP-47L1GTF: /2520090089_OSSP/2520090089_Practical/Practical-07$ cat /proc/767/maps
5fae0084b000-5fae0084c000 r--p 00000000 08:30 13860 /home/uday/2520090089_OSSP/2520090089_Practical/Practical-07/memory
5fae0084c000-5fae0084d000 r-xp 00001000 08:30 13860 /home/uday/2520090089_OSSP/2520090089_Practical/Practical-07/memory
5fae0084d000-5fae0084e000 r--p 00002000 08:30 13860 /home/uday/2520090089_OSSP/2520090089_Practical/Practical-07/memory
5fae0084e000-5fae0084f000 r--p 00002000 08:30 13860 /home/uday/2520090089_OSSP/2520090089_Practical/Practical-07/memory
5fae0084f000-5fae00850000 rw-p 00003000 08:30 13860 /home/uday/2520090089_OSSP/2520090089_Practical/Practical-07/memory
5fae31738000-5fae31739000 rw-p 00000000 00:00 0 [heap]
769f30000000-769f30028000 r--p 00000000 08:30 32948 /usr/lib/x86_64-linux-gnu/libc.so.6
769f30028000-769f301c0000 r-xp 00028000 08:30 32948 /usr/lib/x86_64-linux-gnu/libc.so.6
769f301c0000-769f3020e000 r--p 001c0000 08:30 32948 /usr/lib/x86_64-linux-gnu/libc.so.6
769f3020e000-769f30212000 r--p 0020e000 08:30 32948 /usr/lib/x86_64-linux-gnu/libc.so.6
769f30212000-769f30214000 rw-p 00212000 08:30 32948 /usr/lib/x86_64-linux-gnu/libc.so.6
769f30214000-769f30221000 rw-p 00000000 00:00 0
769f30378000-769f3037b000 rw-p 00000000 00:00 0
769f30381000-769f30383000 rw-p 00000000 00:00 0
769f30383000-769f30387000 r--p 00000000 00:00 0 [vvar]
769f30387000-769f30389000 r--p 00000000 00:00 0 [vvar_vclock]
769f30389000-769f3038b000 r-xp 00000000 00:00 0 [vdso]
769f3038b000-769f3038c000 r--p 00000000 08:30 32929 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
769f3038c000-769f303bb000 r-xp 00001000 08:30 32929 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
769f303bb000-769f303c0000 r--p 00030000 08:30 32929 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
769f303c0000-769f303c8000 r--p 0003b000 08:30 32929 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
769f303c8000-769f303c9000 rw-p 0003d000 08:30 32929 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
769f303c9000-769f303ca000 rw-p 00000000 00:00 0
7ffded743000-7ffded764000 rw-p 00000000 00:00 0 [stack]
uday@DESKTOP-47L1GTF: /2520090089_OSSP/2520090089_Practical/Practical-07$
```

```

uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical/Practical-07$ pmap 831
831:  ./memory
00005d1451002000      4K r---- memory
00005d1451003000      4K r-x-- memory
00005d1451004000      4K r---- memory
00005d1451005000      4K r---- memory
00005d1451006000      4K rw--- memory
00005d148210d000    132K rw--- [ anon ]
000070f431c00000    160K r---- libc.so.6
000070f431c28000   1632K r-x-- libc.so.6
000070f431dc0000    312K r---- libc.so.6
000070f431e0e000     16K r---- libc.so.6
000070f431e12000      8K rw--- libc.so.6
000070f431e14000     52K rw--- [ anon ]
000070f431f6d000     12K rw--- [ anon ]
000070f431f76000      8K rw--- [ anon ]
000070f431f78000     16K r---- [ anon ]
000070f431f7c000      8K r---- [ anon ]
000070f431f7e000      8K r-x-- [ anon ]
000070f431f80000      4K r---- ld-linux-x86-64.so.2
000070f431f81000    188K r-x-- ld-linux-x86-64.so.2
000070f431fb0000    44K r---- ld-linux-x86-64.so.2
000070f431fbb000      8K r---- ld-linux-x86-64.so.2
000070f431fbd000      4K rw--- ld-linux-x86-64.so.2
000070f431fbe000      4K rw--- [ anon ]
00007fff8c873d000   132K rw--- [ stack ]
total                2768K
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical/Practical-07$

```

```

uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical/Practical-07$ cat /proc/856/status | grep -E "VmSize|VmRSS|VmData|VmStk|VmExe|VmLib"
VmSize:    2768 kB
VmRSS:     1936 kB
VmData:     224 kB
VmStk:      132 kB
VmExe:        4 kB
VmLib:     1828 kB
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical/Practical-07$

```

```

uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical/Practical-07$ cat /proc/896/maps | grep -E "\[heap\]|\[stack\]"
5a245d6c0000-5a245d6e1000 rw-p 00000000 00:00 0 [heap]
7ffe53fa9000-7ffe53fca000 rw-p 00000000 00:00 0 [stack]
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical/Practical-07$

```